



Chemistry Ch 3.1: Atomic Models

Chapter 3, Atomic Structure · kitab ke page 27 se 44

Dalton (1803), chaar postulates

1. Atom **na toray ja sakne wala** hai. 2. Ek hi element ke atoms **bilkul ek jaise** hain. 3. Atoms **sadah tanasub** mein jurte hain. 4. Atoms **na banaye ja sakte hain na khatam kiye**.

Rutherford ka gold foil experiment

Polonium se nikalne wale **alpha particles** ko **0.00004 cm** moti sone ki warq par maara.

Teen nataij: atom ka zyadatar hissa **khaali** hai; nucleus **chhota, kasif aur positive** hai; electrons nucleus ke gird ghoomte hain.

Rutherford ka model nakaam kyun hua

Ghoomta hua electron **accelerating charge** hai, aur usay **musalsal energy** chhorni chahiye thi. Phir wo **continuous spectrum** deta, jo kabhi dekha hi nahin gaya. Ye defect yaad rakhein, sawal banta hai.

Bohr (1913), paanch postulates

1. Electrons **daairawi orbits** mein ghoomte hain. 2. Energy nucleus se **faasle ke mutanasib**. 3. Angular momentum **quantized** hai ($h/2\pi$). 4. Orbit badalte waqt roshni **jazb ya kharij**. 5. $\Delta E = E_2 - E_1$.

Quantum Mechanical Model, aur teen naam

Ab **orbit** nahin, **orbital** hai: wo ilaqa jahan electron **milne ka imkaan** sab se zyada ho.

de Broglie (1924): electron ki **dohri fitrat**, lehar bhi aur zarra bhi.

Heisenberg (1927): electron ki **jagah aur momentum** dono ek saath theek maloom nahin ho sakte.

Davisson aur Germer (1927): tajurbe se electron ki **lehar** wali fitrat sabit ki.

Quiz

1. Describe Rutherford's gold foil experiment and the three conclusions he drew. Then state the defect that made his planetary model fail.
2. Write Bohr's five postulates, and name the three scientists behind the quantum mechanical model with what each contributed.